

SHARED AGENT BRIEF — Spain Master's Sweep

Read this fully before your first search. Your slice instruction is in your own prompt.

Candidate

Tunisian citizen (**non-EU**, no prior Spanish residency). Software Engineering degree, SMU MedTech Tunisia, **300 ECTS**, 5-year, completed Aug 2026. Arabic native, French fluent, English fluent, **Spanish A0/A1 now** (B1-B2 possible by Sept 2027). Also a working music producer/DJ. **Target intake: September 2027**. Funding availability is a primary ranking factor, not a footnote.

Two consequences you must internalise:

1. 300 ECTS **exceeds** the usual 240-ECTS entry bar. For every programme, check for advanced standing, *reconocimiento de creditos*, or exemption from *complementos de formacion*. This is a ranked field, never skip it.
2. Language of instruction is a **filter**. Spanish-only programmes are in scope but flagged. Never infer English from an English version of the webpage.

The nine fields (path codes)

- A Sound & Music Computing / Audio Technology
- B Music Information Retrieval / AI for Music (often a TRACK inside a bigger master - record tracks too)
- C Audio Signal Processing & DSP (usually a track inside telecommunications engineering masters)
- AA Artificial Intelligence / Machine Learning
- AB Data Science
- AC Cloud / DevOps / SRE / Distributed Systems (rarely standalone - check software eng, computer eng, cybersecurity masters for cloud/infra tracks, and cert+master hybrids)
- X Business Analytics
- P Music Industry / Label & Publishing Management
- S Events, Festival & Live Entertainment Management

NON-NEGOTIABLE RULES

1. **Primary sources only.** The programme's own page on the university's own domain. Aggregators (Masterstudies, Studyportals, Educaedu, Emagister, FindAMasters, mastermania, quecursar) may be used **to DISCOVER**, never to state a fact. Discovery via aggregator **MUST** be followed by verification on the university site.
2. **Every factual field carries a source URL and an access date.** No URL, no field.
3. **If a fact cannot be found write NOT FOUND.** Never estimate, never infer, never carry a figure over from a similar programme. **NOT FOUND** is a useful result. A plausible invention is a corrupted dataset and a failed task.
4. **Fee figures must state the year they apply to:** EUR X,XXX (2026-27 rates). Spanish public fees are set annually per autonomous community and change.
5. **Non-EU surcharge must be checked explicitly.** Several communities charge non-EU / non-resident students differently. Do not assume the headline price applies.

6. **master universitario vs titulo propio - the single most consequential distinction.**

- **Master universitario** (OFFICIAL): ANECA-verified, appears in RUCT, grants PhD access, EU-recognised, eligible for public scholarships.

- **Titulo propio / master propio**: the university's own certificate. NOT ANECA-verified, **NO PhD access**, weaker recognition, usually ineligible for public funding, and frequently marketed identically to the official kind.

- Verify against RUCT: <https://www.educacion.gob.es/ruct/> If it is not in RUCT it is not official, whatever the marketing says. RUCT is a form-based app that is hard to deep-link; practical route: (a) most official programme pages state their own `codigo RUCT / codigo del titulo` - capture it; (b) google `site:educacion.gob.es/ruct <programme>`; (c) an official master's page virtually always carries the words **Master Universitario** plus a BOE publication reference and an ANECA `/verificacion` link. Absence of all of these = treat as **titulo propio** and say so.

7. **Never trust a translated page for language of instruction.** Find the explicit statement

(`idioma de imparticion / llengua de docencia`). If the English page says English but the official plan says castellano, **the official plan wins**.

8. **Record contradictions, do not resolve them silently.** Two pages disagree -> log both values with both URLs and flag CONFLICT.

WRITE AS YOU GO — THE SINGLE MOST IMPORTANT RULE IN THIS FILE

Append each record to your output file the moment you finish it. Never batch your writes to the end of the task.

This is not style advice. Session limits have twice killed a dozen agents at once, and the ones that died on the line "*now I'll write the records*" produced **nothing at all** — hours of correct, sourced research lost, because it existed only in their heads. The agents that had been appending as they went lost only their current record.

Concretely:

- finish programme 1 → append line 1 → finish programme 2 → append line 2 → ...
- never hold more than ONE finished record unwritten;
- if you are cut off, everything already appended survives and the orchestrator can see exactly where you stopped.

An agent that returns a beautiful summary and an empty file has failed the task completely.

An agent that writes 9 of 15 records and dies has done 60% of a job. Be the second one.

Output discipline

- Write your results **to the JSONL file named in your prompt**, one JSON object per line.
- Return to the orchestrator only a **SHORT** summary: counts, the file you wrote, and anything that blocked you. Do not paste your dataset back.
- Be economical with fetches: search first, fetch the pages that matter. Do not fetch a page twice. Prefer the university's own domain.

Final instruction

An incomplete dataset that is honest about its gaps is a SUCCESS.

A complete-looking dataset containing one invented fee, deadline, or eligibility ruling is a FAILURE, because it will be acted on.

RUCT IS MACHINE-QUERYABLE — use the script, do not guess

`.masters-search/spain/ruct.py` drives the RUCT form directly. Use it instead of guessing official status or trying to fetch the form by URL.

```
python3 .masters-search/spain/ruct.py "inteligencia artificial" "ciencia de datos"
python3 .masters-search/spain/ruct.py --universidad "Pompeu Fabra"
python3 .masters-search/spain/ruct.py --json "sonido"          # machine-readable
```

Columns: RUCT code · title · university · estado (BOE state).

Three traps this script already handles, which will silently corrupt your results if you write your own query instead:

1. **An accented term returns ZERO ROWS, not an error.** `musica` finds 41 titles; `música` finds none. That looks exactly like "no such programme is registered" — the most dangerous wrong answer available here. The script strips accents for you.
2. **The form keeps the previous query in server-side session state.** Reusing one HTTP session makes every term after the first return the FIRST term's results, which looks like a successful query. The script opens a fresh session per term.
3. **RUCT matches substrings, so SHORT terms win.** `tecnologias del sonido` → 2 rows; `sonido` → 3; `musica` → 41. Query short and broad, then filter yourself.

Reading estado matters as much as presence. `TITULACIÓN A EXTINGUIR / EXTINGUIDA` means the title is being wound down and **may not accept a September 2027 intake at all** — record that, because a programme's mere presence in the register is not the same as it being open to you. `Publicado en B.O.E.` is the healthy state.

Citable source for a fee/status claim: RUCT has no deep links, so cite the programme page's own stated `código RUCT / BOE` reference as the primary source, and use this script as the cross-check. Put the RUCT code in `ruct_code` either way.

The backbone is a keyword sweep, so absence from it is NOT a verdict. It was built from 52 term families; a title whose name uses none of them is simply missing. A real case: UCJC's *People Analytics* master (RUCT 3500166) is genuinely official but absent, because no query term matched "people analytics" or "talento". **If a programme is not in the backbone, query RUCT directly with `ruct.py` before concluding anything** — and if that also returns nothing, say "not found in RUCT by these terms", not "not official".

A pre-built backbone of every registered master in the nine fields is at `output/ruct-backbone.jsonl` (fields: `ruct_code`, title, university, estado, active, `path_codes`, `query_terms`). Check it before running your own query — it may already answer you.

Domains that block automated fetching — what to do

Some institution sites sit behind a Cloudflare JS challenge and return **HTTP 403 to every automated fetch** (WebFetch, curl, and a real headless browser alike). Confirmed blocked: **www.upf.edu**, **unir.net**, **il3.ub.edu**, **pointblankmusicschool.com**, **www.threepoints.com** (times out, zero bytes -- a network failure, not a 403), **dks.digital/KSchool**, **www.uclm.es**, **fundacionsgae.org**, **www.techtitute.com** (Cloudflare managed challenge, 403 to WebFetch and to curl with a browser UA).

CORRECTION — **imep.es** is **NOT blocked**. It was on this list in error; it serves curl with a browser UA normally.

The 202-then-200 pattern is commoner than it looks, and it goes both ways. **ucjc.edu** and **urjc.es** return a small shell first and the full body on retry. **nuclio.school** does the reverse — the FIRST curl with a browser UA returned 200 and 285KB, the RETRY a 193-byte shell. **Always try twice before recording a block, and treat a small body as a challenge rather than an answer.** **iqs.edu** needs `curl --cacert /root/.ccr/ca-bundle.crt`.

A blocked main domain does not mean a blocked institution — check subdomains first. Two verified cases: **www.uclm.es** genuinely 403s, but **esi.uclm.es** (which hosts the Big Data master's own microsite) and **preinscripcionmu.apps.uclm.es** (the official preinscription portal, with the full calendar and per-title credits and modality) both serve normally. Likewise **www.unir.net** is blocked while **static.unir.net** serves the official brochures, and **www.upf.edu** is blocked while **bsm.upf.edu** is not. Before writing off a domain, try the faculty, school, campus and application subdomains. **unirioja.es** additionally rate-limits curl to 429 while WebFetch gets through — if one client fails, try the other before recording a block.

CORRECTION — **ucjc.edu** and **urjc.es** are **NOT blocked**. An earlier version of this brief listed **ucjc.edu** as blocked on the strength of an HTTP 202 shell. That is wrong and it cost real coverage. Both show a **202-then-200 pattern**: the first request returns a small shell, and a **retry with a browser User-Agent returns 200 with the full body**. WebFetch returns empty for them; curl with a UA works. **Retry before you record a block** — a 202 shell is a challenge, not a wall.

CORRECTION — **www.unir.net** **IS reachable**. Earlier passes recorded it as blocked on the strength of a 403 to WebFetch and to curl with a bare User-Agent. It returns **200 to curl with a COMPLETE browser header set** — the `Sec-Ch-Ua-*` and `Sec-Fetch-*` headers are what get through; a UA alone is not enough. Any UNIR fact tagged `source_access: search index` should be re-sourced directly.

CORRECTION — **UNIR DOES run a September intake**. This brief previously said it had none, and that was wrong: "Inicio 28 sep 2026" appears on its IA and Ciberseguridad pages, and four of its calendar PDFs are named "Septiembre 2026". UNIR names each cohort by month with a PER code and publishes a full-year calendar at `static.unir.net/CalendariosST/Nac/{Actual,Siguiente}/<slug>.pdf` — use those, they are definitive.

Still true: static.unir.net serves the brochures — UNIR's official programme brochures are served there and fetch fine, which makes ECTS, modality, plan de estudios and access rules primary-sourced rather than index-sourced. Try it before giving up on UNIR.

UNIR publishes no price for any master, anywhere. Its own brochure states the fee *"se calcula en función del total de créditos ECTS matriculados, la forma de pago y las características específicas de cada alumno"*, disclosed only by an adviser after preinscripción. So **NOT FOUND** is the correct and final answer for UNIR tuition — **any UNIR fee figure that appears without a non-UNIR primary source is unsupported and must not be recorded.** Aggregators quote numbers for these; they are not sourceable.

UNIR also runs no September intake — its convocatorias are staggered through the year. For a September-2027 target that is a structural mismatch, not a missing date.

Two UPF specifics, verified — do not over- or under-apply the block.

`mtg.upf.edu` looks like a way round it but **302-redirects into `www.upf.edu`** and hits the same wall, so the Music Technology Group's own pages are genuinely unreachable. But **`bsm.upf.edu` is NOT blocked** — UPF Barcelona School of Management is a separate site serving both WebFetch and curl normally. Do not skip it because of the `upf.edu` warning: its Data Analytics for Business master is official (RUCT 4318274) and its next intake is **27 September 2027**, this candidate's exact target intake.

`fundacionindra.org` and `fundacionaccenture.org` fail with a 502 at the network gateway — that is an environment block, NOT evidence that no scheme exists. Never write those up as a negative finding.

`guiadocent.upf.edu` and `estudiospropios.unizar.es` are additionally refused by the network gateway. Chromium/Playwright does NOT get through — do not spend time on it. The Wayback Machine is rate-limited here and is not a reliable fallback either.

UPF is the single most important institution in this sweep, so this matters. The route that does work:

1. **WebSearch still surfaces the page content** from the search index even when the page cannot be fetched. Use it, and quote what it gives you.
2. **RUCT gives you official status and the RUCT code** regardless of the site being blocked.
3. **Public fees do not come from the university page anyway** — they come from the autonomous community's annual `decreto de precios públicos`, which is fetchable. Use it.

Then be honest about provenance. Any fact obtained from a search index rather than the page itself must be recorded with `"source_access": "search index – page returns 403, not directly fetched"` alongside the URL, and the programme must be listed in `gaps.md` as **needing a human to open it in a normal browser**. Do NOT present indexed content as if you had read the primary page, and do NOT let a blocked domain become an invented fact.

Budget your searches — several wave-1 agents ran out

WebSearch is capped **per agent** (~200 calls) and multiple agents hit the ceiling **mid-slice**, losing their last planned sweeps. The exhaustion message reads as though the limit were session-wide and already drained by others — it is not, and it has been verified that search keeps working elsewhere after an agent reports this. **Do not treat it as a reason to stop early or to skip searching at the start of your slice.** It is your own budget; spend it deliberately:

1. **Check `output/ruct-backbone.jsonl` FIRST.** It already holds every RUCT-registered Spanish master across all nine fields (1,056 titles, 561 currently active) with code,

title, university and estado. Grepping it costs nothing and answers most official-status questions outright — do not spend a search or a RUCT query on something already sitting in that file.

2. Search to **locate** a page; fetch to **read** it. Do not search for facts a page you already have will tell you.
 3. Never fetch the same URL twice — note what you learned the first time.
 4. If you are running low, **write what you have to your JSONL before continuing**. A file with 20 solid records beats an agent that dies holding 40 in its head.
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CORRECTION TO RULE 6 — "not in RUCT" does NOT always mean "not official"

Rule 6 says: *if it is not in RUCT it is not official*. That is true for **university** titles and false for one whole category, and applying it naively misclassifies real official programmes as `título propio`.

Máster en Enseñanzas Artísticas is a separate official register that RUCT does not contain. This was confirmed empirically during wave 1: querying RUCT for the conservatory sector returns zero master records, while those same programmes are ANECA-evaluated and homologated by BOE order (e.g. Orden EFP/673/2023 for the Valencia *Sonología Aplicada y Creación Sonora*). Conservatories — **ESMUC, Musikene, Conservatori del Liceu**, and the superior conservatories of music generally — award in this register.

So there are **three** categories, not two:

Category	In RUCT?	PhD access	Public funding
---	---	---	---
Máster Universitario	yes	yes	yes
Máster en Enseñanzas Artísticas	no — and still official	check per case	check per case
Título propio / Máster de Formación Permanente	no	no	usually no

For an Enseñanzas Artísticas programme, evidence of officiality is **the BOE homologation order**, not a RUCT row. Record the BOE reference in `ruct_code` prefixed `BOE:` and say in notes that RUCT absence is expected here and is not evidence against it.

The real filter is the admission test, not the degree bar — and the degree bar is GENEROUS. An earlier version of this brief said these programmes typically require a `Título Superior de Música`. That was too pessimistic and is now corrected from source: **art. 15 RD 1614/2009 admits the holder of ANY official Grado**, not only a music title. The real gate is **art. 16, the centre-set prueba de acceso** — an audition or portfolio that each centre defines for itself.

So never ask "does it demand a `Título Superior de Música`?" Ask "**what is the prueba de acceso, and can a producer's portfolio satisfy it?**" The distinction decides real cases: one register programme (Musikene's *Mediación, Gestión y Difusión Musical*) selects on merit with **no audition at all** and states it is aimed at graduates from other disciplines; others ask for recent work samples with **no scores required**, which a producer can plausibly meet; only some demand `partitura + audio` or a live audition, which genuinely closes them.

Note also the evaluation chain: the community proposes, **ANECA or a regional agency** evaluates (Aragón's went through ACPUA), the Ministry homologates by BOE order. So "no ANECA report" is **not** evidence against officiality here either.

Note also: **Máster de Formación Permanente / Formación Continua is the post-RD 822/2021 legal name for a título propio**. It reads official. It is not.

Because of all this, `output/ruct-backbone.jsonl` covers the university register only and **does not list conservatory-sector masters** — absence from the backbone is not a verdict.

RUCT detail pages ARE deep-linkable — this fixes rule 2 for official status

The RUCT *search* has no deep links, but the *detail* page does, and it is citable:

```
https://www.educacion.gob.es/ruct/estudio.action?
codigoCiclo=SC&codigoTipo=M&CodigoEstudio=<CODE>&actual=estudios
```

So **every official-status claim can and must carry a real URL** — use this one, with the programme's RUCT code. Put it in `ruct_url`.

That page also answers several ★ schema fields authoritatively, straight from the register:

RUCT detail field	Schema field it answers
Nº Créditos Complementos Formativos	complementary_credits_required — the 300-ECTS question. 0 means no bridging credits are built into the title.
Créditos Totales + the obligatorios/optativos/prácticas/TFM breakdown	ects
Nivel MECES, Nivel académico (RD 822/2021 etc.)	official_status
Fecha de verificación, BOE publication dates, accreditation renewals	official-status evidence with a citable date
Centros en los que se imparte	which campus actually teaches it

`output/ruct-detail.jsonl` already holds all of this for every active registered title in the nine fields. Read it before fetching anything — it is the cheapest source of truth you have.

complementos formativos = 0 in RUCT is not the whole answer. It means the *title* carries no compulsory bridging credits; a university can still impose complementos on an individual applicant at admission based on their background. Record the RUCT figure as evidence, then check the programme's own `acceso y admisión` page for applicant-level rules, and say which of the two you are quoting.

ruct_candidate_code is a HINT, not a fact — challenge it

Batch inputs carry a `ruct_candidate_code` attached by a fuzzy title-and-university matcher. It has already been wrong in a way that would have corrupted a record: it attached Universidad Europea de **Madrid**'s Big Data master to RUCT 3500344, which is Universidad

Europea de **Valencia**'s separate title of the same Spanish name. The real code, printed on UEM's own page, is 4315398.

Spanish universities run near-identical titles at sibling campuses and under bilingual double names, so title overlap alone cannot identify a title. **Confirm the code from the programme's own page (Código RUCT:) or from the RUCT detail page, and say in notes how you confirmed it.** If it does not match the hint, record the correct one and flag the mismatch — do not quietly accept the hint.

The full record schema (wave 2 fills this)

6. Data schema

Every programme record. Fields marked ★ are the ones that decide the outcome — never leave them unsourced.

```
{
  "id": "slug",
  "path_codes": ["A"],           // may match multiple fields
  "programme_name_es": "",
  "programme_name_en": "",
  "institution": "",
  "institution_type": "",        // public | private | specialist school | conservatory
  "city": "",
  "autonomous_community": "",

  "official_status": "",         // ★ máster universitario | título propio | UNKNOWN
  "ruct_code": "",               // ★ empty means NOT official
  "ruct_url": "",

  "ects": 0,                     // 60 | 90 | 120
  "duration_years": 0,
  "modality": "",                // presencial | semipresencial | online
  "language_of_instruction": "", // ★ verbatim from official plan
  "language_source_url": "",
  "language_requirement": "",    // e.g. B2 English certificate required?

  "tuition_total_eur": 0,        // ★ full programme, not per credit
  "tuition_per_ects_eur": 0,
  "tuition_year_of_rates": "",   // ★ e.g. "2026-27"
  "non_eu_surcharge": "",        // ★ yes/no + detail
  "additional_fees_eur": 0,      // enrolment, insurance, admin
  "tuition_source_url": "",

  "entry_requirements": "",
  "accepts_engineering_background": "", // ★ explicit or inferred? say which
  "complementary_credits_required": "", // ★ complementos de formación
  "credit_recognition_available": "",   // ★ the 300 ECTS question
  "credit_recognition_source_url": "",

  "application_window_2027": "", // ★ open and close dates
  "application_rounds": [],
  "non_eu_early_round_advised": "",
  "admissions_contact_email": "",
  "deadline_source_url": "",

  "curriculum_summary": "",
```



```

"notable_faculty_or_lab": "",
"industry_links": "",
"thesis_or_internship": "",

"scholarships_internal": [],      // university's own, with URLs
"scholarship_ids_external": [],    // FK into funding.jsonl

"verification_status": "",        // VERIFIED | CONFLICT | UNCONFIRMED
"conflicts": [],
"sources": [ {"field": "", "url": "", "accessed": "YYYY-MM-DD"} ]
}

```

Notes on filling it

- `complementary_credits_required`: RUCT publishes this per title. **null in output/ruct-detail.jsonl means the register does not publish the row at all — that is NOT the same as 0.** Write NOT PUBLISHED IN RUCT for null, 0 (RUCT) for zero, and add whatever the programme's own admission page says on top, labelled as such.
 - `tuition_total_eur`: the FULL programme, not per credit. If the page gives €/ECTS, multiply and show your arithmetic in notes (e.g. "82 EUR/ECTS x 60 = 4,920").
 - `non_eu_surcharge`: check explicitly. **Confirmed real at UAM, UPC, UC3M, UB, UVic, UAH and URJC** (URJC publishes a distinct `Estudiantes extracomunitarios, no residentes` row at 84.07 EUR/credit vs 45.02 first enrolment). If the page is silent, that is NOT FOUND, not no.
- The decisive legal point, confirmed from Catalonia's Decret 96/2026 art. 1.5:** a Spanish long-stay study authorisation does **not** confer resident status — *"l'autoritzacio d'estada de llarga durada per estudis ... no equival a la condicio de residents"*. So this candidate is legally non-resident for fee purposes and cannot escape the surcharge by enrolling and living there. Several decrees (Catalonia, Asturias) do not fix the non-EU price at all — they authorise each university to set it up to 100% of cost, so the real figure must be read off **that university's own tariff**, never the decree.
- `application_window_2027`: the Sept-2027 intake. If only the 2026-27 calendar is published, record THAT with its year clearly marked and say the 2027-28 dates are not yet out — do not shift dates forward by a year yourself.
 - `verification_status`: leave as PENDING; wave 4 sets it.